

Lab 3: Component Modelling & Architectural Pattern Selection

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Problem Statement: #25 — Municipal Solid Waste Route Optimizer

We chose Layered Architecture for the Municipal Solid Waste Route Optimizer System.

Architectural Choice

A three-tier Layered Architecture — Presentation, Business, and Data — as shown in the accompanying component diagram. The Presentation layer holds the Supervisor Dashboard, Driver Mobile App, and Citizen Portal components; the Business layer holds the Route Optimization Engine, Bin Status & Sensor Ingestion Service, and Complaint Management Service; the Data layer holds the shared Bin & Route Database.

Two Reasons

Reason 1 — UI diversity, one shared rule set: Three distinct, independently-evolving client surfaces sit on top of one shared rule set. The Supervisor Dashboard, Driver Mobile App, and Citizen Portal each serve a different actor with a different interaction style, but all three ultimately depend on the same route/bin logic (FR-001, FR-003, FR-005). Layering lets any one of these UIs change or be rebuilt without touching the Route Optimization Engine that NFR-001's 45-second recomputation target depends on.

Reason 2 — one shared, tightly-coupled data model: Every functional requirement ultimately reads or writes one consistent data model — bin fill state, route history, and complaints (exactly the grouping Lab 2's four epics already revealed). FR-002's sensor updates and FR-003's driver mark-as-collected updates both mutate the same bin record, so keeping all business services against one authoritative Data layer avoids the cross-service consistency problems a microservices split would introduce for data this tightly coupled.

Security Advantage

No Presentation-layer component is allowed to touch the Data layer directly — each must go through a named Business-layer interface (e.g. CollectionUpdate, ComplaintSubmission). This means the Driver Mobile App can never write bin state on its own; every write is validated (driver identity, bin ID, complaint ownership) inside the Business layer before it reaches the database, centralising authentication and audit logging at one boundary instead of duplicating access checks across three separate client apps.

Performance Benefit

Because the Presentation layer only calls into the Business layer through well-defined interfaces and never blocks on the Data layer directly, the Supervisor Dashboard stays responsive while the Route Optimization Engine performs its NFR-001-mandated recomputation (under 45 seconds for 50 trucks across 10,000 bins). The heavy computation is isolated to one layer, so it can be optimised independently (indexing, caching, batched writes in the Data layer) without any change to client-facing code.